





What value will come in place of the question mark (?) in the following question?  
निम्नलिखित प्रश्न में प्रश्न चिह्न (?) के स्थान पर क्या मान आएगा?

$\frac{2}{8} \leftarrow \frac{5}{8} \leftarrow \frac{3+1}{8} = \frac{3}{8} + \frac{1}{8} = ?$

- A. 0.4
- ☒ B. 0.5
- C. 0.6
- D. 0.2
- E. 1

What value will come in place of the question mark (?) in the following question?

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$$\left(\frac{5 \times 1}{6 \times 1}\right) + \left(\frac{2 \times 2}{3 \times 2}\right) + \left(\frac{1 \times 3}{2 \times 3}\right)$$

$$\frac{5}{6} + \frac{4}{6} + \frac{3}{6}$$

$$\frac{12}{6}$$

$$\Rightarrow 2$$

$$\frac{5}{6} + \frac{2}{3} + \frac{1}{2} = ?$$

Handwritten calculations in red:

- $83.33\%$  (circled)
- $66.66\%$  (circled)
- $50\%$  (circled)
- $200\%$  (circled)

$$2 \leftarrow \frac{12}{6} \leftarrow \frac{5+4+3}{6}$$

- A. 4
- B. 0.5
- C. 0.75
- ☒ D. 2
- E. 1

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$$\frac{8}{2} = \frac{1}{4}$$
$$x = 2$$

$$1 - \frac{3}{4}$$

$$\frac{(4-3)}{4}$$

$$\frac{1}{4}$$

$$1 - \frac{4}{17} = \frac{13}{17}$$

$$1 - \frac{9}{13} = \frac{4}{13}$$

$$1 - \frac{5}{7} = \frac{2}{7}$$

$$\frac{3}{4} + \frac{?}{8} = 1$$

- A. 4
- B. 5
- C. 6
- D. 2 ✓
- E. 10

$$1 - \frac{1}{3} = \frac{2}{3}$$

$$1 - \frac{7}{8} = \frac{1}{8}$$

$$1 - \frac{2}{7} = \frac{5}{7}$$

$$1 - \frac{2}{9} = \frac{7}{9}$$

$$1 - \frac{3}{5} = \frac{2}{5}$$

$$1 - \frac{9}{10} = \frac{1}{10}$$

$$1 - \frac{4}{5} = \frac{1}{5}$$

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$$\left( ? = \frac{5}{6} \right)$$

$$? \times \frac{5}{6} = \frac{25}{36}$$

- A.  $\frac{4}{3}$
- B.  $\frac{3}{2}$
- ✓ C.  $\frac{5}{6}$
- D.  $\frac{1}{2}$
- E.  $\frac{6}{5}$

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$$\frac{1}{8} - \frac{5}{8} = \frac{?}{4}$$


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$$2 \frac{2}{8} = \left( \frac{?}{4} \right)$$

$1 = ?$

$$\frac{7}{8} - \frac{?}{4} = \frac{5}{8}$$

- A. 2
- B. 22
- C. 123
- ☒ D. 1
- E. None of these

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$$\left( \frac{5}{8} - \frac{3}{8} \right) = \left( \frac{8}{x} \right)$$

$$\left( \frac{5}{8} - \frac{3}{8} \right) = \frac{8}{x}$$

$$x = 32$$

$$\frac{5}{8} - \frac{8}{?} = \frac{3}{8}$$

- A. 2
- B. 4
- C. 16
- ☒ D. 32
- E. None of these

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$$\frac{7}{8} - \frac{1}{4} - \frac{1}{4} = \frac{8}{x}$$

$$\frac{7 - 2 - 2}{8} = \frac{8}{x}$$

$$\frac{3}{8} = \frac{8}{x}$$

$$3x = 64$$

$$x = \frac{64}{3}$$

$$\frac{7}{8} - \left( \frac{8}{x} \right) - \frac{1}{4} = \left( \frac{1}{4} \right)$$

$$\frac{7}{8} - \left( \frac{8}{?} + \frac{1}{4} \right) = \frac{1}{4}$$

- ☒ A.  $64/3$
- B.  $32/3$
- C.  $64/7$
- D.  $32/7$
- E. None of these

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$$\frac{5}{9} - \frac{8}{3} + \frac{1}{9} = \frac{5}{x}$$

$$\frac{5}{9} - \left( \frac{8}{3} - \frac{1}{9} \right) = \frac{5}{?}$$

$$\frac{5 - 24 + 1}{9}$$

$$\frac{+6 - 24}{9}$$

$$\frac{-18}{9} = \frac{5}{x}$$

$$-2 = \frac{5}{x}$$
$$\left( x = -\frac{5}{2} \right)$$

$$x = -2.5$$

- A. 2
- B. - 2
- C. 2.50
- ☒ D. - 2.50
- E. None of these

Types of Fractions:

$\frac{p}{q} \rightarrow$  Numerator  $\rightarrow 3121$   
 $\frac{p}{q} \rightarrow$  Denominator  $\rightarrow 22$

### 1.) Proper fraction

When  $N < D$

$\Rightarrow \frac{1}{4}, \frac{3}{7}, \frac{9}{13}, \frac{8}{17}, \frac{1}{2}$

Arrows indicate:  $\frac{1}{4}$  (N < D),  $\frac{3}{7}$  (N < D),  $\frac{9}{13}$  (N < D),  $\frac{8}{17}$  (N < D),  $\frac{1}{2}$  (N < D).

### 4.) Unlike fraction

Comparison

When Numerators are same, But denominators are different

$\frac{2}{3}, \frac{2}{5}, \frac{2}{7}, \frac{2}{9}$

### 2.) Improper fraction

When  $N > D$

$\left(\frac{8}{7}\right), \left(\frac{9}{4}\right), \left(\frac{14}{5}\right), \left(\frac{17}{9}\right), \left(\frac{3}{2}\right)$

1 + Proper fraction

### 5.) Like fraction

When denominators are same, but Numerators are different

$\frac{5}{7}, \frac{1}{7}, \frac{8}{7}, \frac{9}{7}, \frac{2}{7}$

### 7.) Equivalent fraction

$\left[\frac{2}{3} = \frac{4}{6} = \frac{50}{150} = \frac{70}{140}\right]$

### 3.) Mixed fraction

Integer Part      fractional Part

$1\frac{1}{4}, 2\frac{3}{4}, 7\frac{9}{13}, 8\frac{2}{7}$

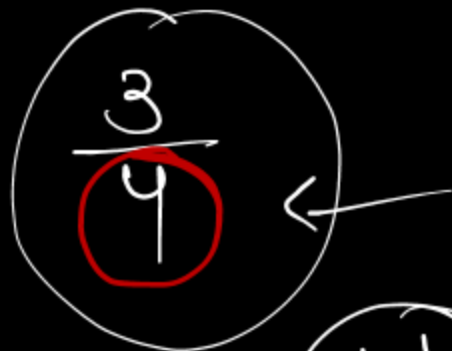
### 6.) Unit fraction or unique fraction

When  $N = 1$

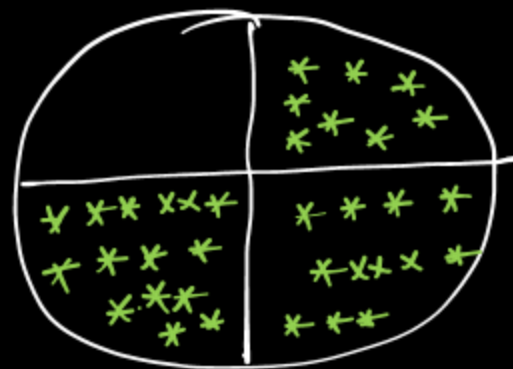
$\frac{1}{2}, \frac{1}{7}, \frac{1}{9}, \frac{1}{10}$

$\left(\frac{2}{7} = \frac{4}{14} = \frac{14}{49}\right)$

152211



$$\Rightarrow 1\frac{1}{7}$$



$$(1) + (\frac{3}{7}) = \frac{10}{7}$$

$$\frac{8}{7}$$

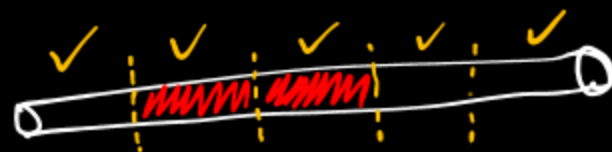
$$\Rightarrow \frac{7+1}{7}$$

$$\Rightarrow \frac{7}{7} + \frac{1}{7}$$

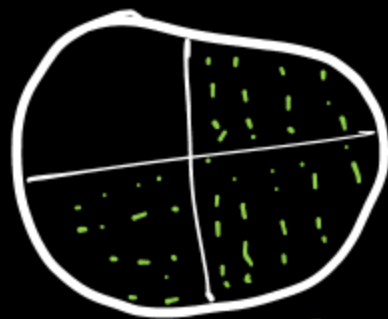
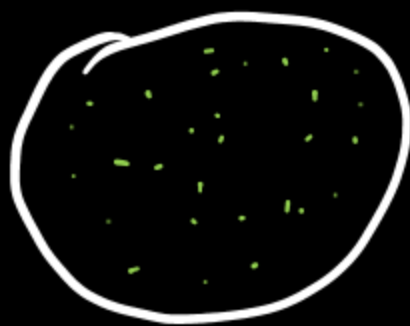
$$\Rightarrow 1 + \frac{1}{7}$$

$$\Rightarrow \frac{1}{7}$$

$$\frac{2}{5}$$



$$\frac{1}{7} + \frac{1}{7} = \frac{2}{7}$$



$$1 + \frac{3}{4} = \frac{7}{4}$$

$$7 \frac{1}{4}$$

$$2 \frac{1}{3}$$

$$5 \frac{2}{9}$$

$$\begin{array}{r} 7 \\ 4 \overline{) 29} \\ \underline{28} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \\ 3 \overline{) 7} \\ \underline{6} \\ 1 \end{array}$$

$$\begin{array}{r} 5 \\ 9 \overline{) 47} \\ \underline{45} \\ 2 \end{array}$$

$$\begin{array}{r} 9 \\ 15 \overline{) 143} \\ \underline{135} \\ 8 \end{array}$$

$$\begin{aligned} \frac{29}{4} &= \frac{28+1}{4} \\ &= \frac{28}{4} + \frac{1}{4} \\ &= 7 + \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \frac{7}{3} &= \frac{6+1}{3} \\ &= 2 + \frac{1}{3} \end{aligned}$$

$$\left( 2 + \frac{1}{3} \right)$$

$$\begin{aligned} \frac{47}{9} &= \frac{45+2}{9} \\ &= 5 + \frac{2}{9} \end{aligned}$$

$$\left( 5 + \frac{2}{9} \right)$$

$$\begin{aligned} \frac{143}{15} &= \frac{135+8}{15} \\ &= 9 + \frac{8}{15} \end{aligned}$$

$$\begin{array}{r} 7 \\ 4 \overline{) 29} \\ \underline{28} \\ 1 \end{array} \rightarrow \left( 7 \right) + \left( \frac{1}{4} \right)$$

$$\begin{array}{r} 2 \\ 3 \overline{) 7} \\ \underline{6} \\ 1 \end{array} \rightarrow 2 + \frac{1}{3}$$

$$\begin{array}{r} 5 \\ 9 \overline{) 47} \\ \underline{45} \\ 2 \end{array} \rightarrow 5 + \frac{2}{9}$$

$$\begin{array}{r} 9 \\ 15 \overline{) 143} \\ \underline{135} \\ 8 \end{array} \rightarrow 9 + \frac{8}{15}$$

$$9 \frac{8}{15}$$

$$9 + \frac{8}{15}$$



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$$\textcircled{3} + \textcircled{\frac{3}{4}} + \textcircled{4} + \textcircled{\frac{1}{2}}$$

Handwritten annotations: 0.75 above  $\frac{3}{4}$ , 0.50 above  $\frac{1}{2}$ . Arrows point from the decimal values to the fractions.

$$3 + \frac{3}{4} + 4 + \frac{1}{2}$$

$$\begin{cases} * 0.75 \\ * 0.50 \\ * 1.25 \end{cases}$$

$$3\frac{3}{4} + 4\frac{1}{2} = ?$$

$$\textcircled{7} + 1.25 = 8.25$$

$$= 7 + 1 + 0.25 = \textcircled{8.25}$$

- ☒ A. 8.25
- ☐ B. 8
- ☐ C. 12.25
- ☐ D. 2
- ☐ E. 10

$$\left(\frac{3}{4}\right) + \left(\frac{1}{2}\right) = \frac{3+2}{4} = \frac{5}{4}$$

$$\frac{4+1}{4}$$

$$\left(\frac{4}{4}\right) + \left(\frac{1}{4}\right)$$

$$\left(1 + \frac{1}{4}\right)$$

$$1 + 0.25$$

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$$\overset{\checkmark}{2} + \frac{5}{6} + \overset{\checkmark}{3} + \frac{2}{3} + \overset{\checkmark}{4} + \frac{1}{2}$$

$$\overset{\checkmark}{9} + \frac{5}{6} + \frac{2}{3} + \frac{1}{2}$$

$$\overset{\checkmark}{9} + \frac{(5+4+3)}{6}$$

$$9 + \frac{(12)}{(6)}$$

$$9 + (2) = 11 \checkmark$$

$$2\frac{5}{6} + 3\frac{2}{3} + 4\frac{1}{2} = ?$$

- A. 7
- B. 8
- C. 9
- D. 10
- ☒ E. 11

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$$[2 - 5 + 4]$$

$$1 + \left\{ \frac{1}{9} - \frac{1}{3} + \frac{1}{3} \right\}$$

$$\left( 1 + \frac{7}{9} \right)$$

$$\left( \frac{16}{9} \right)$$

$$+ \left( 2 + \frac{7}{9} \right) - \left( 5 + \frac{1}{3} \right) + \left( 4 + \frac{1}{3} \right) = ?$$

- A.  $\frac{4}{3}$
- ~~B.  $\frac{16}{9}$~~  ✓
- C.  $\frac{25}{16}$
- D.  $\frac{25}{9}$
- E. None of these

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$$\left(\frac{13}{4}\right) + \left(\frac{x}{8}\right) = (5)$$

$$\frac{x}{8} = \left(5 - \frac{13}{4}\right)$$

$$\frac{x}{8} = \left(\frac{7}{4}\right)$$

$$x = 14$$

$$\left(3\frac{1}{4}\right) + \left[\frac{?}{8}\right] = 5$$

- A. 12
- B. 14 ✓
- C. 16
- D. 18
- E. 22

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$$? \times \left( \frac{5}{6} \right) = \left( \frac{35}{6} \right)$$

$? = 7$  ✓

$$? \times \frac{5}{6} = 5\frac{5}{6}$$

- A. 5
- B. 6
- C. 7 ✓
- D. 8
- E. None of these

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$$\frac{\left(\frac{33}{8}\right)}{\left(1 + \frac{x}{4}\right)} = \left(\frac{5}{8}\right)$$

$$\frac{\left(\frac{33}{8}\right)}{\left(\frac{5}{8}\right)} = \left(1 + \frac{x}{4}\right)$$

$$\left(\frac{33}{5}\right) = \left(1 + \frac{x}{4}\right)$$

$$\frac{33}{5} - 1$$

$$\left(\frac{28}{5}\right) = \left(\frac{x}{4}\right)$$

$$\left(\frac{112}{5}\right) = x = 22.4$$

$$\frac{112}{5} = \frac{112 \times 2}{10} = \frac{224}{10}$$

$$4\frac{1}{8} \div \left[1\frac{?}{4}\right] = \left(\frac{5}{8}\right)$$

- A. 21.50
- B. 22
- ☒ C. 22.40
- D. 25
- E. None of these

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$$\textcircled{5} + \frac{1}{7} \textcircled{-1} - \frac{1}{2} - \frac{5}{7} = \frac{3}{x}$$

$$5 + \frac{1}{7} - \frac{3}{2} - 1 - \frac{1}{2} = \left(\frac{5}{7}\right)$$

$$5\frac{1}{7} - \left(\frac{3}{?} + 1\frac{7}{14}\right) = \frac{5}{7}$$

$\underbrace{1 + \frac{1}{2}}$

$$4 - \frac{15}{14} = \frac{3}{x}$$

$$\frac{56 - 15}{14} = \frac{41}{14} = \frac{3}{x}$$

$$\frac{1}{7} - \frac{1}{2} - \frac{5}{7}$$

$$\frac{2 - 7 - 10}{14}$$

$$1 + \frac{7}{14} \quad \boxed{x = \frac{42}{41}}$$

$$1 + \frac{1}{2}$$

- A. 41/42
- B. 42/41 ✓
- C. 31/2
- D. 2/31
- E. None of these

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$$\frac{1}{7} \ominus \left[ 6 - \frac{4}{7} + 2 \right]$$

$$\frac{1}{7} - \left[ 6 - \left( \frac{8}{14} - \frac{4}{2} \right) \right] = ?$$

$$\frac{1}{7} - \left[ 6 + \frac{4}{7} - 2 \right]$$

$$-8 + \frac{5}{7}$$

$$\frac{5}{7} - 8 = \frac{5 - 56}{7} = -\frac{51}{7}$$

- A.  $-2/7$
- B.  $22/7$
- ~~C.  $-51/7$~~
- D.  $-7$
- E. None of these

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Last

$$\frac{18 + 28}{24} = \frac{46}{24}$$

$$\frac{1}{8} \times \frac{46}{24} = ? \times \frac{23}{96}$$

$$? = 1$$

$$\frac{1}{8} \times \left( \frac{3}{4} \times \frac{7}{6} \right) = ? \times \frac{23}{96}$$

Butter fly

- A. 2
- B. 3
- C. 4
- D. 1 ✓
- E. None of these